

Automatic Soap Making Machines & Production Line

The Process of Manufacturing Soap From Oil and Caustic Soda

The whole process is divided into two major steps.

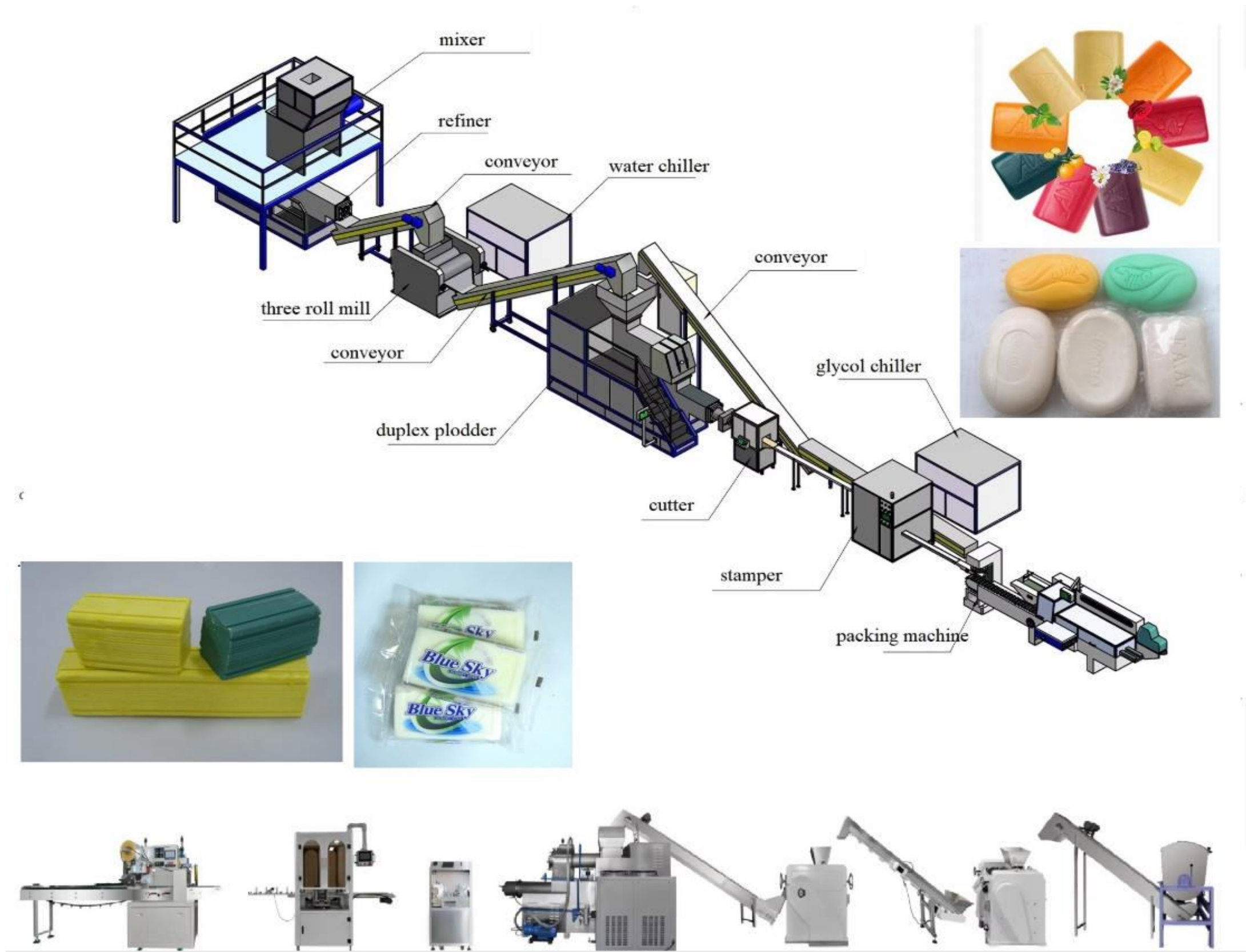
The first step (Part A) is the saponification reaction of oil and caustic soda to form a soap base. This soap base is then processed through a vacuum drying system and a pelletizer, which extrudes it into soap noodles.

The second step (Part B): The soap noodles are put into a soap mixer, which blends the soap noodles with perfume and pigment. The mixture is then fed into a three-roll mill, where the soap noodles are ground into thin soap sheets. These thin sheets are transferred via a conveyor belt to a soap plodder machine, where they are refined again and continuously extruded into a long soap bar with the desired shape. Afterward, the long bar is cut and shaped into individual soap bars of the required size using a soap embossing and cutting machine or a soap stamping machine. Finally, the finished soap bars are wrapped by a packaging machine.

For most beginners with no experience in soap manufacturing and a small budget, we suggest choosing Part B for the initial production.



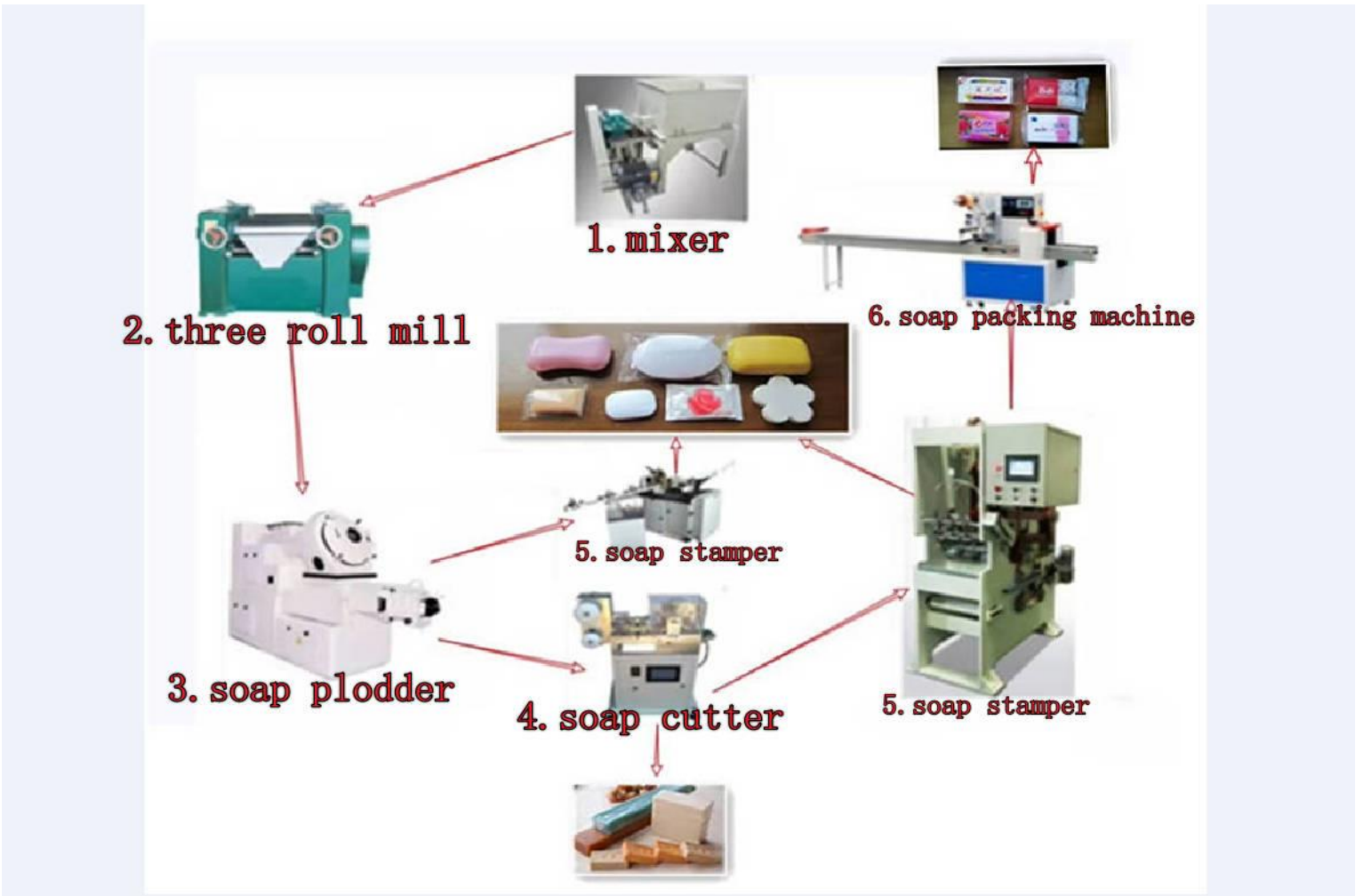
Flow Chart with Soap Noodles to Manufacture Soap

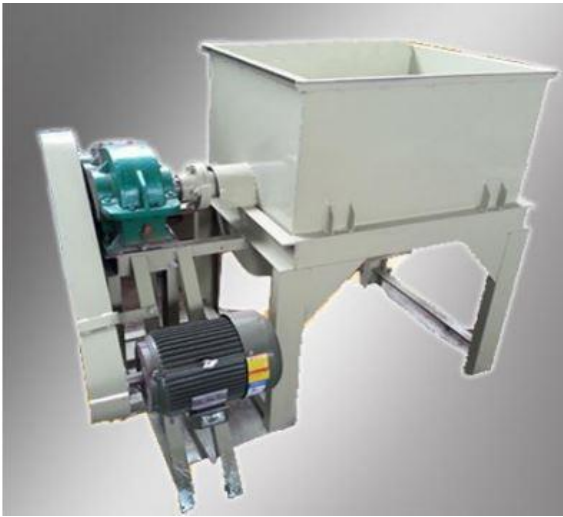
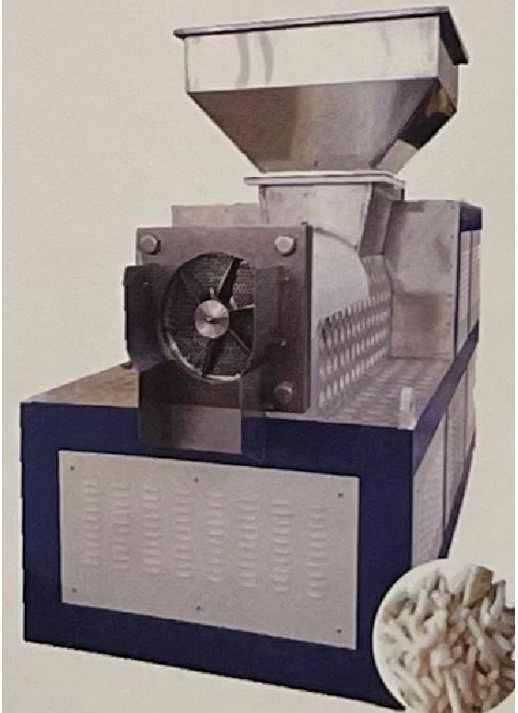


Part A:
Saponification and Vacuum Drying System---Soap Noodles Production Equipment

1		Small medium integrated saponification and vacuum drying system from the different oil and caustic soda to produce soap noodles. Capacity: 50-100KG/hour 300-500KG/hour
2		Large saponification and vacuum drying system from the different oil and caustic soda to produce soap noodles. Capacity: 1000-1500KG/hour 2000-2500KG/hour 3000-5000KG/hour

Part B:
Mixing, Grinding, Extruding, Molding Equipment and Soap Packing Machine---Soap Production Line



3		Industrial Soap Mixer: Core Functions and Working Principle: This equipment is a critical piece of machinery in the front-end stage of the soap production line. It is primarily used to highly efficiently and uniformly blend soap noodles with various additives, such as perfumes, pigments, essential oils, and other auxiliary ingredients. The mixer is equipped with internal screw blades. During operation, the motor drives the main shaft to rotate, utilizing the mechanical motion of the screw to continuously lift, tumble, and scatter the soap noodles from the bottom. This powerful shearing and convective action allows liquid perfumes and solid pigments to rapidly penetrate and evenly coat the soap noodles in a short time. Consequently, it establishes a smooth, refined, and consistently colored soap base foundation for the subsequent milling and plodding (extruding) processes. Structural Features: -Drive and Power System: Equipped with a high-performance electric motor on the left side, the machine smoothly transmits power to the main mixing shaft via pulleys and a gear reducer, providing high-torque mixing and shearing force. -Mixing Tank/Hopper: The top features a large-capacity, open-top charging hopper, making it easy to feed in soap noodles and auxiliary raw materials. -Discharge Design: A manually controlled discharge gate (outlet) is located at the bottom of the mixing tank. Once mixing is complete, it can be manually opened to discharge the uniformly blended materials into a lower container or onto a conveyor belt. Materials and Frame: The entire unit is supported by a robust metal steel frame. The parts in contact with the material are typically made of stainless steel to prevent corrosion from fragrances or chemicals and to ensure production hygiene.
4		Refine Plodder Applications: -Deep Refining and Homogenization: This equipment is primarily used to apply intense mechanical extrusion and shearing to the mixed or milled soap materials. It breaks down the microscopic particles of the soap body further, ensuring that the perfumes, pigments, soap noodles, and auxiliary ingredients are deeply integrated at a molecular level while eliminating internal air bubbles. -Essential for Premium Toilet Soap: As indicated in the original English description, it is chiefly utilized in the production of toilet soap and bar soap. Toilet soap demands exceptionally high standards for texture, smoothness, and foam fineness, making this machine indispensable for increasing the density and compactness of the soap base. A Bridge in the Process Line: Within the production line, it is typically positioned after the three-roll mill and before the soap cutter/stamping machine, taking responsibility for converting loose soap material into a continuous, dense soap bar. Key Features: -Powerful Screw Extrusion System: The internal core consists of a single or twin-screw with a large lead. Through forceful propulsion and enclosed extrusion, it generates immense mechanical shearing force, which forces the physical modification of the soap materials, rendering them more ductile and resilient. -End Refining Mesh/Plate Design: At the front end of the discharge outlet (inside the circular part of the machine head shown in the picture), it is generally equipped with a fine orifice plate and multi-layer stainless steel filter meshes. As the soap material is forced through the filter, any remaining tiny hard particles are thoroughly crushed or intercepted, ensuring the final soap surface is silky smooth with zero graininess.
5		Three Roller Mill Crystal Transformation and Modification: After mixing, the internal crystallization of the soap may be uneven. The three-roll mill utilizes high-pressure extrusion to force the transformation of soap crystals, forming a crystalline structure with better ductility and resilience. This ensures that the final soap is less prone to cracking and does not easily become mushy when soaked in water. Complete Elimination of Graininess: The mixed soap material may contain incompletely dissolved solid pigments, additives, or hard soap chunks. Through the ultimate crushing action of the three rollers, these tiny particles are completely pulverized, ensuring that the soap texture is smooth and refined. Deep Homogenization: Under intense shearing force, perfumes, essential oils, and additives undergo a secondary compounding with the soap base at a microscopic level, rendering the color and fragrance of the entire batch perfectly consistent.

6		Double-Stage Vacuum Soap Plodder (or Vacuum Soap Extruder): -High-Efficiency Vacuum Deaeration: The soap material undergoes complete extraction of air and bubbles within a sealed vacuum chamber. This renders the internal structure of the soap exceptionally dense and compact, effectively preventing internal cavities or cracks while making the final soap bar more durable and less prone to becoming mushy when soaked in water. -High-Pressure Refining and Final Extrusion: Utilizing a two-stage relay screw extrusion design, the equipment thoroughly compacts the loose flakes or granules of soap material. It then forcefully pushes the material through the machine head, converting it into a continuous, uniform, and highly dense soap bar. -Determining Finished Surface Brightness: Through the thermal extrusion process at the final machine head, it imparts an extremely smooth, even, and glossy surface to the extruded soap bar, which directly defines the premium visual quality of the toilet soap or laundry soap.
7		Automatic Soap Stamping Machine / Soap Stamper XD120 Model -Three-Dimensional Compression Molding: The soap emerging from the previous process (the vacuum plodder) is merely a continuous long bar. Through the closure and clamping of precision upper and lower (or left and right) molds, this equipment stamps the soap blanks under high pressure into finished toilet soap bars featuring perfect contours, rounded corners, or specific three-dimensional shapes. -Brand Logo and Pattern Imprinting: Simultaneous with the stamping and molding process, the text or graphics engraved on the molds are deeply imprinted onto the soap surface. This includes brand trademarks, net weight indications (such as 100g), special efficacy symbols, or decorative lines. -Integration of Cutting and Stamping: In small-scale toilet soap production lines, this equipment typically integrates a cutting mechanism. It first cuts the long soap bar precisely into blanks of uniform weight, and then immediately feeds them into the molds for stamping, achieving a seamless connection between the cutting and molding stages.
8		Automatic Toilet soap stamping machine XDA1000type Applications: -Mass Full-Automatic Stamping and Molding: Specially designed for medium to large-scale toilet soap production lines, this equipment possesses full automation capabilities. Its core function is to stamp the cut soap blanks at high speeds and in large batches into finished soap bars with specific three-dimensional shapes, rounded corners, or contours. -Multi-Station Synchronous Imprinting (Multiple Bars per Stroke): The mold structure of this model supports the simultaneous stamping of multiple soap blanks (multi-cavity molds). While stamping and molding, it efficiently imprints brand logos, trademarks, weights, and decorative patterns deeply into the surface of each soap bar. -Premium Hotel and Toilet Soap Production: It is primarily utilized to manufacture various high-quality oval, round, or custom-shaped toilet soaps, bath soaps, and disposable hotel guest soaps, which demand exceptionally high geometric edge smoothness and pattern definition. Key Features: -1000 KG/Hour High-Capacity Design: The XDA1000 model belongs to a high-throughput category. Combined with its multi-cavity mold structure that stamps multiple bars at a time, the machine's production capacity can reach 1000 kg/hour, making it perfect for continuous, large-scale industrial manufacturing. -Exceptional Mold Compatibility and Flexibility: Different specifications of soap molds can be securely fixed and replaced on this machine. Users only need to switch the molds to flexibly alternate between producing different shapes (such as oval, round, square) and various sizes of soap on the same unit. -Requirement of Three Crucial Auxiliary Systems: To guarantee high-speed and stable operation, the manufacturer's specification explicitly indicates that it must be connected to the following auxiliary machinery: a.Low-Temperature Chiller: Circulates chilled cooling water through the molds to freeze them, preventing the soap from sticking during high-frequency, high-pressure stamping and ensuring smooth demolding. b.Air Pump / Air Compressor: Provides pneumatic power for air-assisted demolding, material pushing, or purging. c.Vacuum Pump: Works in tandem with vacuum suction cups or a vacuum chamber to pick up, position, or hold the soap blanks securely during the stamping stroke to prevent displacement.

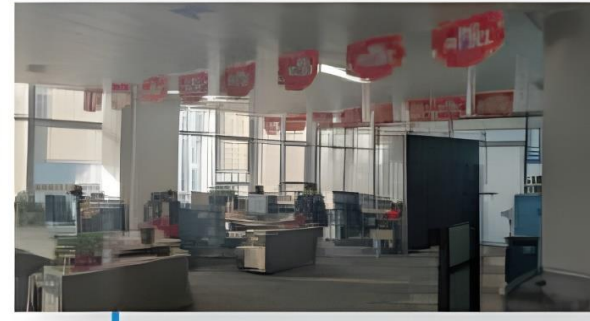
9		Automatic pneumatic toilet soap stamping machine XD100 Applications -Full-Automatic Pneumatic Stamping and Molding: Utilizing the pneumatic thrust provided by an air compressor as its core power source, this equipment replaces traditional manual or heavy mechanical linkages to automatically and highly efficiently press soap blanks into finished toilet soap bars with specific, aesthetically pleasing three-dimensional shapes. -Fine Logo and Information Imprinting: At the exact moment the pneumatic molds close, clear brand trademarks (logos), weights, expiration dates, or decorative patterns are deeply and uniformly imprinted onto either a single side or both sides of the soap bar. -Small-Scale Daily Chemical Production: It specifically targets the automated production needs of small-scale toilet soaps, bath soaps, or disposable hotel guest soaps. It is ideally suited for early-stage startups, laboratory research and development, and production lines characterized by multi-variety, small-batch customization. Auxiliary Requirements: -Air Compressor: As indicated in the manufacturer's description, this machine must be connected to an external air pump. Since all cutting actions, mold downward movements, and material pushing functions throughout the entire unit are driven by pneumatic components, a stable air supply is mandatory to maintain proper pressure. -Low-Temperature Chiller: Chilled water must be circulated through the soap molds during the stamping process to lower their temperature. Freezing the molds causes the soap surface to shrink minutely, which prevents the soap body from adhering to the molds during high-frequency stamping and ensures smooth geometric edges and crisp, smudge-free patterns upon demolding.
10		Automatic Soap Embossing and Cutting Machines -Precision Continuous Cutting: The core responsibility of this equipment is to precisely and rapidly cut the continuous bar of soap extruded from the vacuum plodder into uniform individual soap blocks (blanks) according to the preset target dimensions. -Continuous Surface Embossing: Simultaneous with the forward conveyance and cutting of the soap bar, the machine's front-mounted roller molds continuously roll over the soap surface, smoothly and clearly imprinting (rotary embossing) brand logos, patterns, or text onto the soap body. -Highly Compatible Full-Automatic Operation: Equipped with an intelligent automatic tracking function, the machine's running speed maintains absolute synchronization with the extrusion speed of the preceding plodder, ensuring a high-speed, continuous production line with zero material jamming.
11		Low Temperature Chiller -Precision Deep Mould Cooling: This equipment primarily circulates and delivers low-temperature cooling liquid to provide continuous and forced temperature reduction for the metal moulds inside the automatic soap stamping machine. -Achieving Perfect and Easy Demolding: When ambient or lukewarm soap blanks undergo high-pressure stamping inside the moulds, they are highly susceptible to adhesion caused by friction and compression. By freezing the moulds, the chiller induces a minute thermal shrinkage on the soap surface in contact with the metal, thereby allowing the soap bar to disengage from the mould extremely smoothly and easily. -Defining Finished Visual Quality: Without the deep-cooling protection of a chiller, the soap is highly prone to mould sticking, smudging, or edge damage during the demolding process. The chiller ensures that every finished soap bar leaving the line possesses a mirror-like bright surface and a crisp, clear logo.
12		Cooling water machine -Frictional Heat Dissipation to Prevent Soap Melting: As the three-roll mill tears soap at high speeds and the vacuum plodder extrudes soap under heavy duty and high pressure, a substantial amount of physical frictional heat is generated. This equipment injects circulating chilled water into the hollow rollers of the mill and the barrels/jackets of the plodder to forcefully carry away heat, preventing the soap from softening, melting, or sticking to the rollers due to overheating. -Preserving Volatile Fragrances and Heat-Sensitive Components: The perfumes, natural essential oils, and special active ingredients added to soap are mostly heat-sensitive substances that easily volatilize or deteriorate at elevated temperatures. The cooling water machine strictly locks the processing temperature within an acceptable ambient or slightly cool range, perfectly preserving the fragrance and efficacy of the soap. -Ensuring Extrusion Consistency and Structure: During the vacuum extrusion process, if the barrel temperature rises too high, the soap may deform or develop a loose internal structure because of localized surface melting. Through precise cooling water temperature control, it guarantees that the extruded soap bar maintains a highly uniform and compact internal texture.

13		Vacuum Pump -High-Efficiency Chamber Vacuum Extraction: The core responsibility of this equipment is to connect with the sealed chamber of the vacuum plodder, completely and forcefully extracting the air and bubbles trapped between the soap granules inside. -Moisture Removal from Materials: While establishing a high-vacuum environment, the negative pressure generated by the vacuum accelerates the volatilization of excess trace moisture on the soap surface and pulls it out. This fine-tunes the wet-to-dry consistency of the material, enabling the soap bar to meet standard hardness and compactness. -Maximizing Finished Internal Density: By thoroughly exhaustin g air and moisture via the vacuum pump, it effectively eliminates the occurrence of tiny internal cavities or cracks within the finished soap bar. This fundamentally enhances the durability of the soap, preventing it from crumbling or breaking apart easily when soaked in water.
14		Mini integrated soap production equipment -One-Stop Micro Soap Production Line:this is a micro soap production line designed specifically for home workshops, artisans, small studios, or educational laboratories. Integrated Multi-Station Refining and Plodding: Despite its compact size, it seamlessly integrates the entire front-end core process onto a single compact frame, covering everything from raw material feeding (soap noodles and additives) and mixing to intensive refining and final high-pressure soap bar extrusion. -Scalable Upgrade for Premium Artisanal and Custom Soaps: It is primarily used to help artisanal workshops upgrade from low-efficiency traditional cold-process or manual mold-pouring methods. By utilizing an industrial extrusion process, it efficiently manufactures high-quality custom soaps with a much denser texture and excellent crack resistance. Key Features Disruptive Modular and Highly Integrated Design: It compactly fuses the physical structures of multiple independent machines from traditional production lines—such as the mixer, refiner, and plodder—onto a single steel base frame. Featuring an extremely small footprint, it can easily fit into a standard room or a small workshop. -Effortless Single-Person Operation and Maintenance: The entire unit is equipped with a unified, integrated electrical control box (the clearly visible control cabinet on the right). With all processing modules operating in linked synchronization, it drastically simplifies the tedious piping connections and material conveyance between multiple individual machines, allowing one single person to complete the entire process from raw material feeding to final extrusion. -Extremely Low Energy Consumption and Startup Capital Costs: The energy consumption design of this model has been thoroughly optimized for micro and home environments, making it deliverable via standard household electricity or low-power commercial electricity. While strictly guaranteeing an industrial-grade soap texture, it reduces initial hardware investment and plant rental costs to the absolute minimum.
15		Soap Packing Machine,pillow style High-Speed Full-Automatic Pillow-Style Packaging: The core responsibility of this equipment is to automatically wrap the molded solid soap bars into plastic composite film, executing horizontal sealing and cutting at both ends and longitudinal sealing at the back to form a fully sealed, pillow-style package. Waterproof, Moisture-Proof, and Anti-Volatilization Protection: Soap is highly susceptible to environmental factors; excessive dryness leads to cracking, while excess moisture causes softening due to dampness, and fragrances easily volatilize. Through the airtight film wrapping of this machine, it effectively locks in moisture and aroma, extending the shelf life of the finished products. Enhancing Commercial Appearance and Branding: Supporting various types of color-printed films, the equipment precisely aligns and displays patterns, barcodes, and production dates on the front and back of the packaging bag, making the finished products ready for direct delivery to supermarkets or hotel terminals for retail.



16		Cartoning Machine / Soap Packing Machine,cardboard box style Full-Automatic Carton Forming and Cartoning: The core responsibility of this equipment is to automatically suction flat, pre-fabricated cardboard blanks, erect them into formed cartons, precisely push the soap bars inside, and automatically complete the carton lid folding, tucking, or hot-melt glue sealing. Premium Branding and High-End Soap Packaging: Cardboard box packaging is generally utilized for retail presentation of medium-to-high-end toilet soaps and beauty bars (such as essential oil soaps, anti-mite soaps, and baby soaps). The rigid cardboard provides superior crush protection and allows for exquisite printing graphics, enhancing the premium brand image. Synchronized Feeding of Inserts or Accessories: Certain advanced configurations support automatic, synchronized feeding of product manuals, promotional leaflets, or moisture-absorbing desiccant sachets together with the soap bar, achieving multi-component integrated packaging. 
17		Soap Packing Machine, soft paper tye Double-Layer Soft Paper Automatic Wrapping and Tucking: The core responsibility of this equipment is to apply a two-layer wrap to the molded soap bars. The internal layer is typically a stiff paperboard stiffener/liner (used for shape retention and structural support), while the outer layer is a coated soft paper or composite paper printed with exquisite graphics. During packaging, it automatically executes end-folding and corner-tucking, forming a classic, retro wrapped-bar appearance. Standard for Traditional Retail Bar Soaps: This packaging style is the iconic signature of globally renowned soap brands like Safeguard, Lux, and Bee & Flower. Compared to plastic film, soft paper wrapping delivers a more eco-friendly texture and superior breathability, making it widely utilized for mass-produced commercial toilet soaps and facial cleansing bars. High-Efficiency Hot-Melt Glue Sealing: Upon completion of the outer soft paper folding, the equipment precisely applies a small amount of hot-melt glue to the tucked corners via an integrated glue-spraying system for adhesive sealing. This ensures a secure, non-loosening package while maintaining effortless tear-open convenience for consumers. 
18		soap mould will be customized as per buyer required soap shape,size and logo Imparting the Final Three-Dimensional Geometric Form: The mould consists of two primary mating halves (the upper male die and lower female die). When the soap blank is fed inside and subjected to high-pressure mechanical or pneumatic stamping, the soap body undergoes plastic deformation within the mould cavities, giving rise to perfect oval, round, rectangular, or various bespoke irregular geometries. In-Depth Presentation of Brand Visual Identity: The center of the cavity typically features high-precision CNC engraving to etch the inverted brand logos, registered trademarks, net weight indications (such as 125g), or refined decorative pinstripes. During the stamping stroke, these alphanumeric characters and graphics are sharply and legibly compressed onto either a single side or both sides of the soap bar. -Fulfilling Market Diversification and Flexible Production: As stated in the original English description, the moulds support comprehensive customization. By switching different moulds on the exact same stamping press, manufacturers can flexibly transition production lines within mere tens of minutes to produce soaps of entirely distinct sizes, dimensions, shapes, and branding, flawlessly responding to market demands.

Workshop



Certifications



Packing and Shipment

FREIGHT LOGISTICS

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Jasun machinery equipment are sold to 50+ countries around the world



FAQ

Q1:

Do you accept the customization?

All of our machine can be customized as per client's requirement.

Q2: How do you guarantee this machine normally operating?

Every machine is tested by our factory and other client before shipping, We will adjust the machine to optimal effect before delivery. And spare is always available and free for you in warranty year.

Q3: How can I install this machine when it arrives?

We will send the engineers overseas to help client installing, commissioning and training.

Q4: What should I do to choose the best machine for us?

A4: 1) Tell me the material you want to fill, we'll choose the suitable type of machine for you to consider.

2) After choose the suitable type of machine, then tell me the filling capacity you need for the machine.

3) At last tell me the inner diameter of your container to help us choose the best diameter of the filling head for you.

Q5: Do you have manual or operation video for us to know more about the machine?

Yes, we will send you the manual and operation video after you ask for us.

Q6: Do you have manual or operation video for us to know more about the machine?

Yes, we will send you the manual and operation video after you ask for us.

Q7: If there's some spare parts that broke, how to solve the problem?

First of all, please take the picture or make a video to show the problem parts.

After the problem is confirmed by our engineer, we'll send you the spare parts for free within warranty year, but the shipping cost should be paid by your side.